Noah Sun

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Education

University of Waterloo – BASc. in Mechatronics Engineering (Candidate) September 2025 – April 2030

Relevant Coursework: Algorithms and Data Structures, Circuits, Digital Logic GPA: 3.92 / 4.00

Experience

Embedded Flight Software Developer – Waterloo Aerial Robotics Group September 2025 – Current

- Led 4 engineers to develop a **32 MB** flash driver using C++ on the STM32 Nucleo to store config details and crash logs
- Built RTOS-integrated Flash Translation Layer with garbage collection and wear leveling, enhancing flash longevity

V5RC Team Programming Lead – Checkmate Robotics | Demo Video | GitHub June 2019 – August 2025

- Led team of 6 in robot design and library architecture, winning the Create (innovation) Award at VEX Worlds 2024
- Engineered ArkLib, a modular C++ motion control library focused on scalability and flexibility using OOP, qualifying the team to VEX Worlds in **5 consecutive years** and winning the Think (programming) Award at international events
- Designed and tuned PID and control algorithms, achieving **1”/1° precision** movements across **300+ matches**

Vex AI Competition Team Embedded Systems Lead – Mi3L Schools May 2024 – June 2024

- Developed an API for the VEX Brain to process 3D spatial data from the Intel RealSense depth camera via the Jetson Nano for adaptive field navigation, winning the Innovate Award and Skills World Champion at VEX AI Worlds 2024

Forward Deployed Engineer – Ascendance Foundry January 2026 – April 2026

- Built a GTM automation that uses Python Playwright to find leads and generate outreach emails using Claude API
- Developed **5 AI workflow automations**, introducing input validation and reducing human error by **80%**
- Implemented **2 ETL pipelines** using Azure Data Factory, delivering per-project statistic dashboards in Power BI

Co-founder and Lead Developer – Elapse | Google Play | App Store | GitHub June 2024 – August 2025

- Led 4 developers to build an iOS/Android Flutter app serving the **100k+** member VEX Robotics Competition community with real-time schedules, rankings, and match data, receiving **3500+ downloads** and a **5.0-star rating**
- Designed adaptive match times and scouting forms, increasing team productivity at tournaments by **35%**
- Integrated Firebase and Cloud Functions to sync data between team members, reducing scouting times by **30%**

Projects

Bare-Metal UART Command Shell (STM32) | GitHub

- Built a bare-metal command shell for an STM32F446 using only **9.3 KB of flash and 2.4 KB of RAM**, with no heap
- **Reduced flash usage by 32%** by replacing the HAL with register-level drivers for GPIO, USART2, and SysTick
- Verified ring buffer correctness with **12 ASan/UBSan-checked unit tests**, including randomized model-based tests

AutoFleet | GitHub

- Built a multi-robot central arbiter with space-time A* path planning to avoid collisions during coordinated traversal
- Programmed ESP32 WiFi-UART bridge with a TCP server to relay JSON commands, enabling wireless robot control
- Integrated AprilTag localization with EMA-smoothed pose fused with wheel odometry, boosting precision by **30%**

Redact | Live Demo | GitHub

- Built a detect-track pipeline utilizing RetinaFace and OpenCV, achieving **97% accuracy** by detecting every 5th frame with IoU-tracked fill-in, cutting processing times by **80%** to **4 FPS**
- Designed dual-signal scene-cut detection and EMA smoothing to keep blur boxes stable across shot changes

Technical Skills

Languages: C/C++, Python, Java, JavaScript, TypeScript, Dart, MicroPython
Embedded/Hardware: Raspberry Pi, STM32, NVIDIA Jetson, Arduino, UART, SPI, Soldering, SolidWorks, 3D Printing
Robotics/Controls: ROS2, RTOS, PID Control, Motor Control, Sensor Fusion, Kinematics, Foxglove
AI/Software: OpenCV, TensorFlow, FastAPI, Express.js, Node.js, React Native, Flutter, Matplotlib
Data/DevOps: PostgreSQL, MySQL, Microsoft Azure, Firebase, Pandas, Git, Docker, Linux, GitHub Actions, REST APIs